Ethical Intelligence

# Java Full Stack Syllabus

| Module (with Duration) | Sub-Module / Topics |
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| **(Week 1)** | • Three Tier Architecture, HTML & CSS. |
| **(Week 2)** | • HTML & CSS. |
| **ASSIGNMENT** | • An assignment to create a personal portfolio page.<br>• An assignment Landing Page for Restaurant Site. |
| **Java (Week 3)** | • What and Why,<br>• Java Differences from Other Languages<br>• Installation<br>• Code Structure and Execution<br>• Parts of Java Compiler,<br>• Data Types<br>• Variables<br>• Operators,<br>• Type Casting<br>• Arrays |
| **(Week 4)** | • Arrays and String methods<br>• If - else Statements<br>• Loops<br>• Switch-case<br>• Break-continue<br>• Reading Inputs (Scanner Class) |
| **ASSIGNMENT** | • An assignment to create a calculator<br>• HTML + CSS + JAVA Combined **TEST MOCK INTERVIEWS** |
| **(Week 5)** | • Functions/methods<br>• Functions Parameter<br>• Method Overloading<br>• Scope Recursion<br>• Classes and Objects<br>• Creating Classes<br>• Class Attributes and methods |
| **(Week 6)** | • Constructors<br>• Creating Objects<br>• Memory Allocation<br>• Constructor overloading<br>• Modifiers<br>• Encapsulation<br>• Getter and Setter<br>• Abstraction |
| **(Week 7)** | • Inheritance (Both Types)<br>• Polymorphism<br>• Wrapper Classes<br>• Collection framework(Map included) and their methods<br>• Multiple Inheritance using Interfaces<br>• Enums |

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| **TEST+ INTERVIEWS** | **JAVA TEST+ MOCK INTERVIEWS** |
| **(Week 8)** | • Final Classes and Methods<br>• Regex<br>• File handling<br>• Exception Handling(Try-catch-finally)<br>• Custom Exception<br>• Working with Dates |
| **(Week 9)** | • Static & Final Keyword<br>• Nested Classes<br>• Generics<br>• Optional Class<br>• Java 8 Functional Programming |
| **(Week 10)** | • Multithreading and concurrency |
| **REVISION AND ASSIGNMENTS (Week 11)** | **REVISION AND ASSIGNMENTS** |
| **JAVA SCRIPT (Week 12)** | • Datatypes<br>• Operators<br>• Conditional statements<br>• Loops<br>• Functions |
| **(Week 13)** | • Arrays<br>• Objects<br>• Destructuring (arrays/objects)<br>• DOM (Document Object Model)<br>• Error Handling<br>• try...catch...finally |
| **(Week 14)** | • Asynchronous JavaScript<br>• Callbacks<br>• Callback Hell<br>• Promises<br>• Promises chaining<br>• Async/Await<br>• Oops |
| **TEST** | **HTML + CSS + JS Test** |
| **SQL (Week 15)** | • Database and SQL queries.<br>• Database<br>• Tables<br>• Data Types<br>• Create, Insert, Select, Operators,<br>• Where clauses, order by, Alter and Update and Drop<br>• Case When |
| **TEST** | **Basic SQL TEST** |
| **(Week 16)** | • Aggregate Functions<br>• Having Clause<br>• Group by<br>• Joins<br>• Unions<br>• Wild cards constraints (including PK, FK, CK) |

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| | • Stored Procedure |
| **(Week 17)** | • CASE WHEN<br>• views Functions<br>• subquery<br>• indexing trigger<br>• Window Functions normalization |
| **TEST** | **FULL SQL TEST** |
| **(Week 18)** | • SQL and JAVA connections using JDBC |
| **PROJECT** | **A BASIC CRUD-MAJOR PROJECT** |
| **(Week 19)** | • Spring Introduction Dependency<br>• Injection<br>• IoC container<br>• What is Bean<br>• Spring MVC<br>• Spring DAO<br>• Spring AOP |
| **(Week 20 & 21)** | • Introduction of Spring Boot<br>• Annotations<br>• Dependency management using POM.xml<br>• Controller<br>• Services<br>• Models<br>• Creating<br>• REST APIs<br>• Hibernate Introduction<br>• CRUD<br>• Operation Using Hibernate and JPA<br>• Hibernate with Annotation<br>• @Query<br>• Criteria API and etc. |
| **(Week 22)** | • Template Engine (Thymeleaf)<br>• Spring<br>• Security (Authentication) and SMTP |
| **TEST (Week 23)** | **PROJECT, REVISIONS AND TEST** |